Trustworthy Machine Learning

Online Learning

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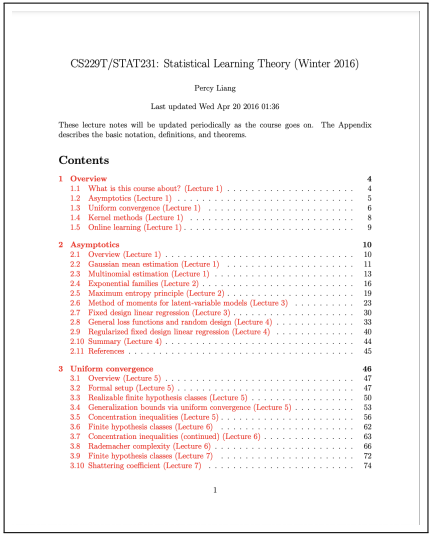
POSTECH

Contents from



CS229T/STAT231: Statistical Learning Theory (Winter 2016)

Percy Liang

Last updated Wed Apr 20 2016 01:36

These lecture notes will be updated periodically as the course goes on. The Appendix describes the basic notation, definitions, and theorems.

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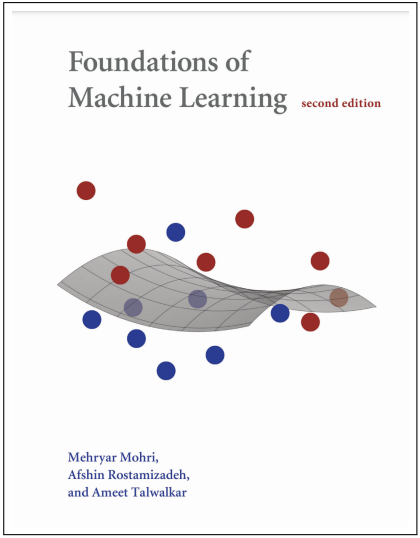


Foundations of Machine Learning **second edition**

Mehryar Mohri,
Afshin Rostamizadeh,
and Ameet Talwalkar

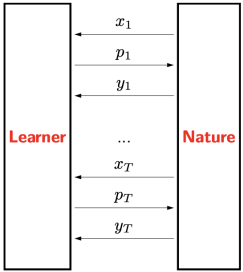
and various papers.

Motivation

- Previously, we consider *statistical learning* (*i.e.*, learning under the i.i.d. assumption)
- However, this assumption can be broken, *e.g.*, distribution shift
- *e.g.*, price data
- Here, we will weaken this assumption.
- batch to online: "how data arrives"
- statistical to adversarial: "how data are generated"

Setup

- prediction task: learn to map an example $x \in \mathcal{X}$ to a label $y \in \mathcal{Y}$
- online learning game between a learner and nature



Protocol:

for $t = 1, \ldots, T$ **do**

Learner receives an example $x_t \in \mathcal{X}$

Learner outputs prediction $p_t \in \mathcal{Y}$

Learner receives a true label $y_t \in \mathcal{Y}$

Learner suffers loss $\ell(y_t, p_t)$

Learner update model parameters

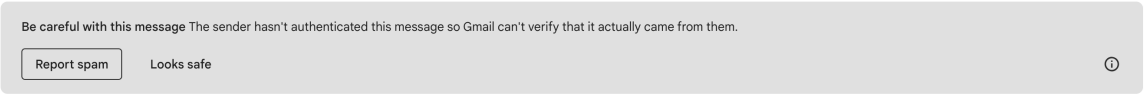
end for

- The learner is a function $\mathcal{A}$ that returns the current prediction given the full history, *i.e.*,

$$p_{t+1} = \mathcal{A}(x_{1:t}, p_{1:t}, y_{1:t}, x_{t+1})$$

[SP: from https://web.stanford.edu/class/cs229t/notes.pdf]

Example: Online Binary Classification for Spam Filtering



- examples: $\mathcal{X} := \{0,1\}^d$ are boolean feature vectors (presence or absence of a word)
- labels: $\mathcal{Y} := \{+1,-1\}$ are whether a document is spam or not
- zero-one loss: $\ell(y_t, p_t) = \mathbb{1}\left(y_t \neq p_t\right)$ is whether the prediction was incorrect

Remarks

- In batch learning, we have a training phase and test phase; but in online learning, they are interleaved.
- The online learning setup allows to use the full history (growing in time by $O(T)$ and in space by $O(T)$). This means that we can use any batch learning algorithm on the history at each time. However, online algorithms tend to be lightweight, *i.e.*, the amount of work by an algorithm should not grow with t.
- Online learning algorithms have the potential to adapt.
- For some applications (*e.g.*, spam filtering), examples are generated by an adversary.

"Goodness" Metric

How to measure the performance of an online learning algorithm?

- the cumulative loss of the learner, *i.e.*,

$$\sum_{t=1}^{T} \ell(y_t, p_t)$$

- No! In the adversarial setting, the adversary can manipulate data to make the learner trivially bad loss.
- What do you do when your grade is awful? Compare to the best grade in your class!

Regret

Definition

$$\text{Regret}(h) := \underbrace{\sum_{t=1}^{T} \ell(y_t, p_t)}_{\text{learner}} - \underbrace{\min_{h \in \mathcal{H}} \sum_{t=1}^{T} \ell(y_t, h(x_t))}_{\text{best expert}}$$

- $\mathcal{H}$ is a class of experts.
- The best export is a role model of the leaner.
- We will consider the worst case regret (*i.e.*, labeled examples are generated by an adversary)

Need Randomization

Setup:

- binary classification, *i.e.*, $y \in \{-1, +1\}$
- zero-one loss, *i.e.*, $\ell(y_t, p_t) \coloneqq \mathbb{1}\left(p_t \neq y_t\right)$
- the learner is fully **deterministic**.

claim

For all learner $\mathcal{A}$, there exists an $\mathcal{H}$ and the sequence of labeled examples such that

$$\textit{Regret} \geq \frac{T}{2}.$$

- Too bad...

Need Randomization: Why? Intuition

- An adversary (who has full knowledge of the leaner) can choose y_t to make it different to the learner's choice p_t.
- Thus, the learner's cumulative loss is T!
- Not yet; how about the best expert's loss?
- Consider two experts, *i.e.*, $\mathcal{H} := \{h_{-1}, h_{+1}\}$ (where h_y always predict y).
- Thus, we have

$$\ell(y_t, h_{-1}(x_t)) + \ell(y_t, h_{+1}(x_t)) = 1 \quad \Rightarrow \quad \sum_{t=1}^{T} \ell(y_t, h_{-1}(x_t)) + \sum_{t=1}^{T} \ell(y_t, h_{+1}(x_t)) = T$$

$$\Rightarrow \sum_{t=1}^{T} \ell(y_t, h_{-1}(x_t)) \leq \frac{T}{2} \quad \text{or} \quad \sum_{t=1}^{T} \ell(y_t, h_{+1}(x_t)) \leq \frac{T}{2}$$

$$\Rightarrow \text{Regret}(h) := \underbrace{\sum_{t=1}^{T} \ell(y_t, p_t)}_{=T} - \underbrace{\min_{h \in \mathcal{H}} \sum_{t=1}^{T} \ell(y_t, h(x_t))}_{\leq \frac{T}{2}} \geq \frac{T}{2}.$$

Add an Assumption without Randomization

Assumption (realizable)

Assume that the best expert $h^* \in \mathcal{H}$ obtains zero cumulative loss (i.e., $\ell(y_t, h^*(x_t) = 0$) for all $t \in \{1, \ldots, T\}$.

- This impose restrictions on adversaries.
- We saw a similar assumption in PAC learning.

Halving Algorithm

Idea:

- Combine opinions of experts.
- Under the realizable assumption, keep only correct experts.

```
HALVING(H)
  H_1 ← H
  for t ← 1 to T do
      RECEIVE(x_t)
      ŷ_t ← MAJORITYVOTE(H_t, x_t)
      RECEIVE(y_t)
      if (ŷ_t ≠ y_t) then
          H_{t+1} ← {c ∈ H_t : c(x_t) = y_t}
      else H_{t+1} ← H_t
  return H_{T+1}
```

- $\mathcal{H}_t$: a set of correct experts.

Halving Algorithm: A Regret Bound

claim

Under the realizable assumption, for any $((x_t, y_t))_{t=1}^T$*, we have*

$$\textit{Regret} \leq \log_2 |\mathcal{H}|.$$

Proof sketch: [SP: check notation]

- Let M be the number of mistakes.
- For each mistake, at least half of the experts are eliminated, *i.e.,* if p_i made a mistake,

$$\frac{V_{i+1}}{V_i} \leq \frac{1}{2} \Rightarrow \frac{|V_{T+1}|}{|\mathcal{H}|} \leq \frac{1}{2^M}.$$

- Due to the realizable assumption, we have

$$1 \leq |V_{T+1}|.$$

- $M =$ Regret.

Halving Algorithm: A Regret Bound

claim

Under the realizable assumption, for any $((x_t, y_t))_{t=1}^{T}$*, we have*

$$\textit{Regret} \leq \log_2 |\mathcal{H}|.$$

- Very strong results due to the realizable assumption.
  - after a finite number of iterations, the predictor never makes mistakes.

Remove the Realizable Assumption

- The realizable assumption is too strong
- Let remove this.
- Then, we need a randomization algorithm.
- One example: Exponential weighing algorithm.

Exponential Weighing Algorithm

- $\mathcal{H}$: a set of experts
- $\ell(\cdot) \in [0, 1]$

Algorithm 1 Exponential Weighting Algorithm

```
w_1 ← (1/|H|, ..., 1/|H|)
for t = 1, ..., T do
    Observe x_t
    Predict ŷ_t = h^{i_t}(x_t), where i_t ~ w_t
    Observe y_t
    Update w_{t+1}(i) ∝ w_t(i) exp{−η ℓ(h^i(x_t), y_t)} for all i ∈ {1, ..., |H|}
end for
```

Exponential Weighing Algorithm: A Regret Bound

Theorem

For any loss function ℓ with the range in $[0,1]$, if $\eta = \frac{8\ln|\mathcal{H}|}{T}$, we have

$$\mathbb{E}\textit{Regret} \leq \sqrt{T\ln|\mathcal{H}|}$$

- No realizable assumption.
- "learnable", *i.e.*, $\frac{\text{Regret}}{T} = \sqrt{\frac{\ln|\mathcal{H}|}{T}}$ with a mild assumption on loss.
- Still we assume a finite set of experts $\mathcal{H}$.

Exponential Weighing Algorithm: A Regret Bound I

Proof sketch

- Let $L_t^i := \sum_{s=1}^t \ell(h_i(x_s), y_s)$ and $W_t := \sum_{i=1}^{|\mathcal{H}|} \exp\{-\eta L_t^i\}$ with $W_0 = |\mathcal{H}|$.
- The lower bound of the "weight difference":

$$\ln \frac{W_T}{W_0} = \ln \sum_{i=1}^{|\mathcal{H}|} \exp\{-\eta L_t^i\} - \ln |\mathcal{H}| \geq \ln \left(\max_{i \in \{1,\ldots,|\mathcal{H}|\}} \exp\{-\eta L_t^i\} \right) - \ln |\mathcal{H}| = -\eta \min_{i \in \{1,\ldots,|\mathcal{H}|\}} L_t^i - \ln |\mathcal{H}|.$$

- The upper bound of the "weight difference":

$$\begin{aligned}
\ln \frac{W_t}{W_{t-1}} &= \ln \frac{\sum_{i=1}^{|\mathcal{H}|} \exp\{-\eta L_t^i\}}{\sum_{i=1}^{|\mathcal{H}|} \exp\{-\eta L_{t-1}^i\}} = \ln \frac{\sum_{i=1}^{|\mathcal{H}|} \exp\{-\eta \ell(h_t^i(x_t), y_t)\} \exp\{-\eta L_{t-1}^i\}}{\sum_{i=1}^{|\mathcal{H}|} \exp\{-\eta L_{t-1}^i\}} \\
&= \ln \mathbb{E}_{i_t \sim w_t} \exp\left\{-\eta \ell(h^{i_t}(x_t), y_t)\right\} \leq -\eta \mathbb{E}_{i_t \sim w_t} \ell(h^{i_t}(x_t), y_t) + \frac{\eta^2}{8} \\
\Rightarrow \ln \frac{W_T}{W_0} &\leq -\eta \sum_{t=1}^T \mathbb{E}_{i_t \sim w_t} \ell(h^{i_t}(x_t), y_t) + \frac{\eta^2 T}{8}
\end{aligned}$$

Exponential Weighing Algorithm: A Regret Bound II

Proof sketch

- For any $s \in \mathbb{R}$ and a random variable $X \in [a, b]$, $\ln \mathbb{E} e^{sX} \leq s\mathbb{E}X + \frac{s^2(b-a)^2}{8}$.

- Combine the lower and upper bounds:

$$-\eta \min_{i \in \{1,\ldots,|\mathcal{H}|\}} L_T^i - \ln |\mathcal{H}| \leq -\eta \sum_{t=1}^{T} \mathbb{E}_{i_t \sim w_t} \ell(h^{i_t}(x_t), y_t) + \frac{\eta^2 T}{8} \Rightarrow$$

$$\sum_{t=1}^{T} \mathbb{E}_{i_t \sim w_t} \ell(h^{i_t}(x_t), y_t) + \min_{i \in \{1,\ldots,|\mathcal{H}|\}} L_T^i \leq \frac{\eta T}{8} + \frac{\ln |\mathcal{H}|}{\eta}$$

Algorithms So Far

- Halving Algorithm
  - Assumption: realizable
  - Assumption: finite $\mathcal{H}$

Algorithms So Far

- Halving Algorithm
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- Exponential Weighting Algorithm
  - Assumption: finite $\mathcal{H}$

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Algorithms So Far

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- What's the next?
  - Remove the assumption on the finiteness of $\mathcal{H}$ (under some assumptions)

Algorithms So Far

- Halving Algorithm
  - Assumption: realizable
  - Assumption: finite $\mathcal{H}$
- Exponential Weighting Algorithm
  - Assumption: finite $\mathcal{H}$
- What's the next?
  - Remove the assumption on the finiteness of $\mathcal{H}$ (under some assumptions)
  - Assumption: realizable

Perceptron Algorithm: History

Perceptron Algorithm: Setup

- $\mathcal{D}$: change over time but require the realizable assumption.
- $\mathcal{H}$: linear functions – additional assumption
- ℓ: 0-1 loss for classification

Perceptron Algorithm

Algorithm 2 Perceptron Algorithm

```
w_1 ← w_0 := 0
for t = 1, ..., T do
    Receives an example x_t ∈ X
    ŷ_t ← sign(w_t · x_t)
    Receives a true label y_t ∈ Y
    if ŷ_t ≠ y_t then
        w_{t+1} ← w_t + y_t x_t
    else
        w_{t+1} ← w_t
    end if
end for
```

Perceptron Algorithm: A Regret Bound

Theorem

Suppose $\|x_t\| \leq r$ *for all* t *and for some* r *and there exists* $\gamma > 0$ *and* $v \in \mathbb{R}^d$ *such that*

$$\gamma \leq \frac{y_t(v \cdot x_t)}{\|v\|}.$$

Then, we have

$$\textit{Regret} \leq \frac{r^2}{\gamma^2}.$$

Perceptron Algorithm: A Proof Sketch

- Let $\mathcal{J} \subseteq \{1, \ldots, T\}$ be the set of time indices when updated.
- Thus, Regret $= |\mathcal{J}|$.
- From the "margin" assumption,

$$\begin{aligned}
\gamma\text{Regret} &\leq \frac{\sum_{t \in \mathcal{J}} y_t (v \cdot x_t)}{\|v\|} = \frac{v \cdot \sum_{t \in \mathcal{J}} y_t x_t}{\|v\|} \\
&\leq \left\| \sum_{t \in \mathcal{J}} y_t x_t \right\| \qquad (1) \\
&= \left\| \sum_{t \in \mathcal{J}} w_{t+1} - w_t \right\| = \|w_{T+1}\| \\
&= \sqrt{\sum_{t \in \mathcal{J}} \|w_{t+1}\|^2 - \|w_t\|^2} = \sqrt{\sum_{t \in \mathcal{J}} \|w_t + y_t x_t\|^2 - \|w_t\|^2} = \sqrt{\sum_{t \in \mathcal{J}} 2 y_t w_t \cdot x_t + \|x_t\|^2} \\
&\leq \sqrt{\sum_{t \in \mathcal{J}} \|x_t\|^2} \leq r \sqrt{\text{Regret}}.
\end{aligned}$$

  - (1): Cauchy-Schwarz inequality, *i.e.*, $u \cdot v \leq \|u\| \|v\|$